

Vishesh Dugar

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SUMMARY

Backend Software Engineer with **2+ years of experience** building distributed, event-driven systems on AWS. Shipped serverless pipelines processing 1M+ daily events, led a Redis migration saving \$64.5K/year, and delivered a zero-downtime inter-cloud migration in one quarter, with recent work in RAG systems and MCP/AgentCore integrations.

TECHNICAL SKILLS

Development Languages: TypeScript, JavaScript, Python, SQL, Bash

Frameworks & Libraries: Node.js, Express.js, FastAPI, Langchain, LangGraph, FastMCP, React

Databases & Caches: Weaviate, PostgreSQL, Redis, Elasticsearch, Aurora, NoSQL (DynamoDB, MongoDB)

Developer Tools: Datadog, Localstack, GitHub Actions

Cloud & DevOps: AWS (ECS, API Gateway, Cognito, AgentCore Bedrock), Docker, Terraform, CI/CD

EXPERIENCE

Software Engineer | ION Group (July 2024 – Present)

- Designed a serverless reporting pipeline on AWS Kinesis Firehose, S3, Glue, and Athena that processes **1M+ daily** API and UI events, generating on-demand customer reports in under **2 minutes** delivered via pre-signed S3 URLs.
- Scaled Elasticache Redis from single node to a clustered architecture, cutting required **RAM 77% (318GB→79GB)** and **infra cost 74% (~\$64.5K/yr savings)** while improving **CPU throughput 3x**.
- Migrated a monolithic admin platform from Azure to AWS with **zero downtime in one quarter**, porting legacy apps to modern stack, transferring 60-70GB of relational data to RDS Postgres via a dual-write sync strategy.
- Architected a serverless, event-driven attachments service (AWS Lambda, SQS, S3, DynamoDB) with React UI enabling users to attach evidence to decisions on subscribed daily monitoring alerts, closing an audit gap with verifiable proof of review reasoning. **3k+ daily requests**.
- Built a resource-level RBAC system from the ground up, covering **200+ APIs** with 5 default roles plus unlimited custom roles, enabling granular, self-service access control across **80,000+ users**.
- Shipped an MCP integration for KYC6 on AgentCore in response to channel partner demand, reducing client agent integration to just **2–3 days** of testing. Adopted by **2 partners serving 20–30 customers** each.
- Independently identified a hiring-coordination bottleneck and shipped a Node.js automation pipeline during a 24-hour hackathon; now adopted in production, **cutting 900 man-hours to 50 for 5,000+ applicants**.

PROJECTS

Paralegal | Node.js, React, AWS, Localstack, Docker, Langchain, Weaviate, OpenAI, Bash | [GitHub](#) | [Demo](#)

- Built a production-grade **RAG** system for legal document querying, with automated ingestion, vector embedding, and a **self-healing feedback pipeline** routing degraded chunks to LLM correction or human review across 10 quality categories.
- Deployed on Hetzner Cloud with a two-tier Terraform architecture emulating full AWS infrastructure via LocalStack, **cutting hosting costs by 95% (\$7 vs \$140/mo)**.

EDUCATION

- Vellore Institute of Technology (**B. Tech in Computer Science**) | GPA: 8.4/10 (September 2020 – July 2024)

AWARDS AND CERTIFICATIONS

[AWS Certified Cloud Practitioner](#) (Valid through May 2027)

[AWS Certified Solutions Architect Associate](#) (Valid through May 2027)

Flipkart GRiD 5.0: Top 1% among 400,000 Participants

MLH: 5x Hackathon Winner along with Sponsor Tracks of Google Cloud, goDaddy, Twilio, Agora